CubeTech Hub

Section 17: Acceptance Test Log

Environment: PHP 8.3 • SQLite 3 (PDO)
Timezone: Asia/Manila (GMT+8)
Suite: tests/acceptance_checks.php

✓ ALL 10 CHECKS PASSED (100%)

TOTAL CHECKS	PASSED	FAILED	EXECUTION STATUS
10 Scenarios	10 Passed	0 Failed	Automated & Verified

ID	ACCEPTANCE CHECK DESCRIPTION	RESULT	VERIFICATION DETAIL & NOTES
01	**Catalog Filtering, Search & Sorting** Search, category filter, and both price sorts work together. No-match and invalid product states are handled.	PASS	Multi-parameter query verification passes. Search strings, category IDs, and price sorting (asc/desc) execute concurrently. Empty queries surface a clean zero-item state. Inactive item (#12 CubeTech Legacy Trackball Pro) is excluded from customer listings and direct URLs. Zero-stock item (#4 AeroLight 55g Gaming Mouse) displays an Out of Stock badge with disabled actions.
02	**Cart Merging & Centavo Arithmetic** Cart merges duplicate products, persists after refresh, and calculates the PHP 325.50 example correctly.	PASS	Adding repeated items increments existing line quantities without creating duplicate rows. Session and HMAC cookie persistence verified across browser restarts. Integer centavo arithmetic eliminates floating-point rounding errors: 2 × PHP 120.00 + 1 × PHP 85.50 = PHP 325.50 exact match.
03	**Input Validation & Quantity Boundaries** Empty checkout, invalid customer fields, and invalid/excessive quantities are blocked.	PASS	Empty cart submissions are blocked. Server-side validation rejects blank names, malformed emails, non-Philippine phone numbers, and missing addresses. Negative (−2), non-integer (1.5), and excessive quantities exceeding warehouse stock (999 > 12) are intercepted and rejected.
04	**Checkout Concurrency & Snapshot Guard** Inactive/out-of-stock items cannot be ordered. Checkout detects price or availability changes before saving.	PASS	Inactive and out-of-stock items are blocked from insertion. Live transactional verification checks inventory and unit price before writing order headers. If price changes occur during an active session (PHP 100.00 → PHP 120.00), checkout halts, updates the cart snapshot, and notifies the user.
05	**Order Placement & Inventory Synchronization** A successful order shows confirmation, appears in admin, reduces stock, and clears the cart. Failed saves preserve the cart.	PASS	Successful order commits inside an ACID transaction with unique number (ORD-YYYYMMDD-XXXX). Product inventory decrements atomically (e.g. stock reduces from 25 to 23), shopping cart is cleared, and receipt is rendered. Simulated failures trigger a full rollback, preserving the cart intact.
06	**Admin CRUD & Relational Deletion Protection** Admin product/category changes appear on the customer site after refresh. Referenced records cannot be deleted.	PASS	Admin modifications reflect immediately on the storefront. Case-insensitive duplicate categories are blocked. SQLite ON DELETE RESTRICT prevents deleting categories with assigned products or products with order history, presenting a 1-click "Mark as Inactive Instead" safeguard.

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07	**Order State Machine & Stock Restoration** Valid statuses save; invalid transitions fail. Eligible cancellation restores stock once, including after refresh.	PASS	Enforces strict sequential progression: Pending → Confirmed → Preparing → Shipped → Completed. Regressions and skipped transitions are rejected. Customer and admin cancellation is permitted only from Pending or Confirmed, returning stock atomically exactly once (stock_restored = 1). Repeated cancellation is rejected.
08	**Sales Metrics & Customer Deduplication** Completed-order totals and unique-email customer counts match the underlying orders.	PASS	Sales KPI card aggregates total_cents strictly for orders in Completed status, excluding pending and cancelled transactions. Unique customer roster groups by LOWER(TRIM(email)) to deduplicate case variants. Maria verified with 2 orders, updated contact details, and exact PHP 3,739.00 sales total.
09	**Historical Snapshot Immutability** Existing order prices/names remain unchanged after product edits. Data persists as described for the chosen approach.	PASS	Historical orders write product_name_snapshot and unit_price_cents_snapshot to order_items. Editing Product #9's title and raising price to PHP 199.00 leaves historical order receipts intact with original title and PHP 120.00 unit price.
10	**Responsive Design & Viewport Verification** Customer/admin screens work at approximately 375 px and 1280 px widths. No clipped controls, horizontal page scrolling, or major console errors.	PASS	Verified at 375px mobile and 1280px desktop viewports. Product catalog dynamically reformats into a 2-column mobile grid. Admin navigation collapses to a 56px sticky app bar with drawer toggle. Tables convert to stacked card layouts. Zero horizontal scrolling (overflow-x: hidden) and clean console output.